KAIST

2026 General Relativity

Homework 3

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September 29, 2026

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#1

For a general metric $g_{\mu\nu}$, the Levi-Civita connection is given by

$$\Gamma^{\lambda}_{\mu\nu} = \frac{1}{2} g^{\lambda\sigma}(\partial_\mu g_{\sigma\nu} + \partial_\nu g_{\sigma\mu} - \partial_\sigma g_{\mu\nu}) \tag{1}$$

If the metric is diagonal, i.e. $g_{\mu\nu} = 0$ for $\mu \neq \nu$, the sum over σ in the above expression reduces to:

$$\Gamma^{\lambda}_{\mu\nu} = \frac{1}{2} g^{\lambda\lambda}(\partial_\mu g_{\lambda\nu} + \partial_\nu g_{\lambda\mu} - \partial_\lambda g_{\mu\nu}) \tag{2}$$

If $\lambda \neq \mu$, $\lambda \neq \nu$, and $\mu \neq \nu$, the Christoffel symbol vanishes.

$$\Gamma^{\lambda}_{\mu\nu} = 0 \tag{3}$$

If $\lambda \neq \mu = \nu$,

$$\Gamma^{\lambda}_{\mu\mu} = -\frac{1}{2} g^{\lambda\lambda}\partial_\lambda g_{\mu\mu} = -\frac{1}{2}(g_{\lambda\lambda})^{-1}\partial_\lambda g_{\mu\mu} \tag{4}$$

If $\lambda = \nu \neq \mu$,

$$\Gamma^{\lambda}_{\mu\lambda} = \frac{1}{2} g^{\lambda\lambda}\partial_\mu g_{\lambda\lambda} = \frac{1}{2}(g_{\lambda\lambda})^{-1}\partial_\mu g_{\lambda\lambda} = \partial_\mu \ln\sqrt{|g_{\lambda\lambda}|} \tag{5}$$

By the non-degeneracy of the metric, $\ln\sqrt{|g_{\lambda\lambda}|}$ is well-defined.

The case $\lambda = \mu \neq \nu$ is not required since Levi-Civita connection is torsion-free, i.e. $\Gamma^{\lambda}_{\mu\nu} = \Gamma^{\lambda}_{\nu\mu}$.

If $\lambda = \mu = \nu$,

$$\Gamma^{\lambda}_{\lambda\lambda} = \frac{1}{2} g^{\lambda\lambda}\partial_\lambda g_{\lambda\lambda} = \frac{1}{2}(g_{\lambda\lambda})^{-1}\partial_\lambda g_{\lambda\lambda} = \partial_\lambda \ln\sqrt{|g_{\lambda\lambda}|} \tag{6}$$

To summarize, the Levi-Civita connection for a diagonal metric is given by

$$\begin{aligned}
&\Gamma^{\lambda}_{\mu\nu} = 0 \\
&\Gamma^{\lambda}_{\mu\mu} = -\frac{1}{2}(g_{\lambda\lambda})^{-1}\partial_\lambda g_{\mu\mu}, \quad (\lambda \neq \mu) \\
&\Gamma^{\lambda}_{\mu\lambda} = \partial_\mu \ln\sqrt{|g_{\lambda\lambda}|}, \quad (\lambda \neq \mu) \\
&\Gamma^{\lambda}_{\lambda\lambda} = \partial_\lambda \ln\sqrt{|g_{\lambda\lambda}|}
\end{aligned} \tag{7}$$

for $\lambda \neq \mu$, $\lambda \neq \nu$, and $\mu \neq \nu$.

#2

Since the given metric is diagonal, using above result, the Christoffel symbols of Levi-Civita connection are

$$\Gamma^{\theta}_{\phi\phi} = -\sin\theta\cos\theta, \qquad \Gamma^{\phi}_{\theta\phi} = \Gamma^{\phi}_{\phi\theta} = \cot\theta \tag{8}$$

and all other components vanish.

Covariant derivative induced by the given connection is

$$\begin{aligned} \nabla_\theta\partial_\theta &= 0 & \nabla_\theta\partial_\phi &= (\cot\theta)\partial_\phi \\ \nabla_\phi\partial_\theta &= (\cot\theta)\partial_\phi & \nabla_\phi\partial_\phi &= -\sin\theta\cos\theta\partial_\theta \end{aligned} \tag{9}$$

(a)

Hence, the velocity vectorfield of meridian ∂_θ is parallel along itself while the velocity vectorfield of parallel ∂_ϕ is not parallel along itself except $\theta = \frac{\pi}{2}$. Therefore, every meridian is a geodesic while the only parallel which is a geodesic is the equator $\theta = \frac{\pi}{2}$.

(b)

Consider a vectorfield $X(\phi) = a(\phi)\partial_\theta + b(\phi)\partial_\phi$ along parallel $\theta = \theta_0$. The condition $\nabla_\phi X = 0$ gives

$$\begin{aligned} \nabla_\phi X &= \nabla_\phi(a(\phi)\partial_\theta + b(\phi)\partial_\phi) \\ &= a'(\phi)\partial_\theta + b'(\phi)\partial_\phi + a(\phi)\nabla_\phi\partial_\theta + b(\phi)\nabla_\phi\partial_\phi \\ &= a'(\phi)\partial_\theta + b'(\phi)\partial_\phi + a(\phi)(\cot\theta_0)\partial_\phi + b(\phi)(-\sin\theta_0\cos\theta_0)\partial_\theta \\ &= (a'(\phi) - b(\phi)\sin\theta_0\cos\theta_0)\partial_\theta + (b'(\phi) + a(\phi)\cot\theta_0)\partial_\phi = 0 \end{aligned} \tag{10}$$

The coordinate expression is

$$\begin{pmatrix} a'(\phi) \\ b'(\phi) \end{pmatrix} = \begin{pmatrix} 0 & \sin\theta_0\cos\theta_0 \\ -\cot\theta_0 & 0 \end{pmatrix} \begin{pmatrix} a(\phi) \\ b(\phi) \end{pmatrix} \tag{11}$$

Hence, the matrix representation of the parallel transport along the parallel $\theta = \theta_0$ is

$$P_{\theta_0}(\phi) = \exp\left(\begin{pmatrix} 0 & \sin\theta_0\cos\theta_0 \\ -\cot\theta_0 & 0 \end{pmatrix}\phi\right) = \begin{pmatrix} \cos(\phi\cos\theta_0) & \sin\theta_0\,\sin(\phi\cos\theta_0) \\ -\dfrac{\sin(\phi\cos\theta_0)}{\sin\theta_0} & \cos(\phi\cos\theta_0) \end{pmatrix} \tag{12}$$

Since we are interested in the parallel transport along the closed curve $\phi \in [0, 2\pi]$, we have

$$P_{\theta_0}(2\pi) = \begin{pmatrix} \cos(2\pi\cos\theta_0) & \sin\theta_0\,\sin(2\pi\cos\theta_0) \\ -\dfrac{\sin(2\pi\cos\theta_0)}{\sin\theta_0} & \cos(2\pi\cos\theta_0) \end{pmatrix} \tag{13}$$

and the resulting vector after the parallel transport when $X(0) = (1, 0)$ is

$$X(2\pi) = P_{\theta_0}(2\pi)X(0) = \begin{pmatrix} \cos(2\pi\cos\theta_0) \\ -\dfrac{\sin(2\pi\cos\theta_0)}{\sin\theta_0} \end{pmatrix} \tag{14}$$

#3

(a)

Let $(\mathcal{M}, g)$ be a pseudo-Riemannian manifold and ∇_μ be the Levi-Civita connection. Let $\gamma : [0,1] \to \mathcal{M}$ be a smooth regular curve and $T = \partial_\lambda$ be the tangent vectorfield along γ. The regularity condition ($T \neq 0$) is necessary to exclude the typical L-shaped counterexample.

Assume there exists a function α along γ such that $T^a \nabla_a T^b = \alpha T^b$. Since T is non-vanishing, at least one component of T is non-zero, $T^\mu \neq 0$. Then,

$$\alpha = \frac{T^a \nabla_a T^\mu}{T^\mu} \tag{15}$$

Since the right-hand side is a smooth function along γ, α is also a smooth function along γ.

Consider a reparametrization $\gamma(t) = \gamma(s(t))$. Then, the tangent vectorfield along $\gamma(s)$ is given by

$$\widetilde{T} = \frac{d\gamma}{ds} = \frac{d\gamma}{dt}\frac{dt}{ds} = \frac{dt}{ds} T \tag{16}$$

Then, the covariant derivative of $\widetilde{T}$ along γ is

$$\begin{aligned} \widetilde{T}^a \nabla_a \widetilde{T}^b &= \left(\frac{dt}{ds}\right)^2 T^a \nabla_a T^b + \frac{d}{ds}\left(\frac{dt}{ds}\right) T^b \\ &= \left(\frac{dt}{ds}\right)^2 (\alpha T^b) + \frac{d^2 t}{ds^2} T^b \\ &= \left(\alpha \left(\frac{dt}{ds}\right)^2 + \frac{d^2 t}{ds^2}\right) T^b \end{aligned} \tag{17}$$

The condition that the right hand side vanishes is equivalent to

$$\frac{d^2 t}{ds^2} + \alpha \left(\frac{dt}{ds}\right)^2 = 0 \iff -\left(\frac{ds}{dt}\right)^{-3} \frac{d^2 s}{dt^2} + \alpha \left(\frac{ds}{dt}\right)^{-2} = 0 \tag{18}$$

$$\iff \frac{d^2 s}{dt^2} = \alpha \frac{ds}{dt} \tag{19}$$

Hence, the reparametrization given by

$$s(t) = \int_0^t \exp\left(\int_0^\tau \alpha(\sigma)\, d\sigma\right) d\tau \tag{20}$$

gives the equation (3.3.1), i.e. $\widetilde{T}^a \nabla_a \widetilde{T}^b = 0$. The reparametrization is well-defined since α is smooth along γ and the integrand is positive, $s(t)$ is strictly increasing and smooth.

(b)

Let t and s be two affine parameters of a geodesic γ, i.e. $T^a \nabla_a T^b = 0$ and $\widetilde{T}^a \nabla_a \widetilde{T}^b = 0$ for $T = \partial_t$ and $\widetilde{T} = \partial_s$. As same as above, the covariant derivative of T along itself is

$$T^a \nabla_a T^b = \left(\frac{ds}{dt}\right)^2 \widetilde{T}^a \nabla_a \widetilde{T}^b + \frac{d^2 s}{dt^2} \widetilde{T}^b = \frac{d^2 s}{dt^2} \widetilde{T}^b \tag{21}$$

Hence, $d^2 s/dt^2 = 0$ which implies $s(t) = at + b$ for some constants a and b.

#4

Since the given metric is diagonal, using above result (7), the Christoffel symbols of Levi-Civita connection are

$$\begin{aligned} \Gamma^r_{\theta\theta} &= -r & \Gamma^r_{\phi\phi} &= -r\sin^2\theta & \Gamma^\theta_{\phi\phi} &= -\cos\theta\sin\theta \\ \Gamma^\theta_{r\theta} = \Gamma^\theta_{\theta r} &= \frac{1}{r} & \Gamma^\phi_{r\phi} = \Gamma^\phi_{\phi r} &= \frac{1}{r} & \Gamma^\phi_{\theta\phi} = \Gamma^\phi_{\phi\theta} &= \cot\theta \end{aligned} \tag{22}$$

other all components vanish.

And the geodesic equation is

$$\begin{aligned} \ddot{r} - r\dot{\theta}^2 - r\sin^2\theta\dot{\phi}^2 &= 0 \\ \ddot{\theta} + \frac{2}{r}\dot{r}\dot{\theta} - \cos\theta\sin\theta\dot{\phi}^2 &= 0 \\ \ddot{\phi} + \frac{2}{r}\dot{r}\dot{\phi} + 2\cot\theta\dot{\theta}\dot{\phi} &= 0 \end{aligned} \tag{23}$$

It is enough to show that this equation of geodesic garuantees that the velocity is constant along the geodesic. The velocity is given by

$$\begin{aligned} v_x &= \frac{d}{dt}(r\sin\theta\cos\phi) = \dot{r}\sin\theta\cos\phi + r\cos\theta\cos\phi\dot{\theta} - r\sin\theta\sin\phi\dot{\phi} \\ v_y &= \frac{d}{dt}(r\sin\theta\sin\phi) = \dot{r}\sin\theta\sin\phi + r\cos\theta\sin\phi\dot{\theta} + r\sin\theta\cos\phi\dot{\phi} \\ v_z &= \frac{d}{dt}(r\cos\theta) = \dot{r}\cos\theta - r\sin\theta\dot{\theta} \end{aligned} \tag{24}$$

and the acceleration is

$$\begin{aligned} a_x &= \ddot{r}\sin\theta\cos\phi + 2\dot{r}\cos\theta\cos\phi\dot{\theta} - 2\dot{r}\sin\theta\sin\phi\dot{\phi} - r\sin\theta\cos\phi\dot{\theta}^2 - 2r\cos\theta\sin\phi\dot{\theta}\dot{\phi} \\ &\quad + r\cos\theta\cos\phi\ddot{\theta} - r\sin\theta\cos\phi\dot{\phi}^2 - r\sin\theta\sin\phi\ddot{\phi} \end{aligned} \tag{25}$$

$$\begin{aligned} a_y &= \ddot{r}\sin\theta\sin\phi + 2\dot{r}\cos\theta\sin\phi\dot{\theta} + 2\dot{r}\sin\theta\cos\phi\dot{\phi} - r\sin\theta\sin\phi\dot{\theta}^2 + 2r\cos\theta\cos\phi\dot{\theta}\dot{\phi} \\ &\quad + r\cos\theta\sin\phi\ddot{\theta} - r\sin\theta\sin\phi\dot{\phi}^2 + r\sin\theta\cos\phi\ddot{\phi} \end{aligned} \tag{26}$$

$$a_z = \ddot{r}\cos\theta - 2\dot{r}\sin\theta\dot{\theta} - r\cos\theta\dot{\theta}^2 - r\sin\theta\ddot{\theta} \tag{27}$$

The equation of geodesic (23) implies that $a_x = a_y = a_z = 0$, which means that the velocity is constant along the geodesic.

Hence, the solution is straight line in $\mathbb{R}^3$.

The detailed calculation of a_x, a_y, a_z is

$$a_x = \sin\theta\cos\phi(\ddot{r} - r\dot{\theta}^2 - r\sin^2\theta\dot{\phi}^2) + \tag{28}$$

$$+ r\cos\theta\cos\phi(\ddot{\theta} + \frac{2}{r}\dot{r}\dot{\theta} - \cos\theta\sin\theta\dot{\phi}^2) \tag{29}$$

$$- r\sin\theta\sin\phi(\ddot{\phi} + \frac{2}{r}\dot{r}\dot{\phi} + 2\cot\theta\dot{\theta}\dot{\phi}) = 0 \tag{30}$$

$$a_y = \sin\theta\sin\phi(\ddot{r} - r\dot{\theta}^2 - r\sin^2\theta\dot{\phi}^2) + \tag{31}$$

$$+ r\cos\theta\sin\phi(\ddot{\theta} + \frac{2}{r}\dot{r}\dot{\theta} - \cos\theta\sin\theta\dot{\phi}^2) \tag{32}$$

$$+ r\sin\theta\cos\phi(\ddot{\phi} + \frac{2}{r}\dot{r}\dot{\phi} + 2\cot\theta\dot{\theta}\dot{\phi}) = 0 \tag{33}$$

$$a_z = \cos\theta(\ddot{r} - r\dot{\theta}^2 - r\sin^2\theta\dot{\phi}^2) - r\sin\theta(\ddot{\theta} + \frac{2}{r}\dot{r}\dot{\theta} - \cos\theta\sin\theta\dot{\phi}^2) = 0 \tag{34}$$